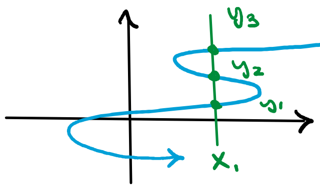
2.3 Functions - 1

Thursday, October 3, 2024 6:38 PM

Def: Let A and B be non-empty sets. A function from A to B is an assignment of exactly one element of B to each element of A.

The function is written $f: A \to B$ and $f(a) = b$ if b is the unique value from B assigned to $a \in A$.

This matches the vertical line test

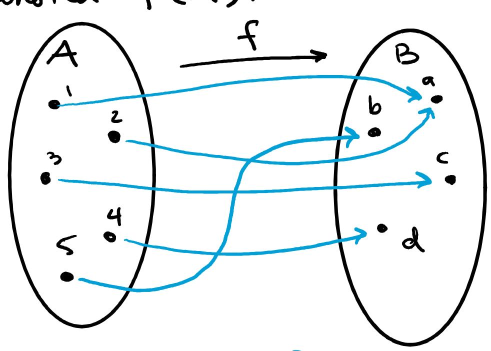


$f(x)$ is not a function because $f(x_1) = y_1$ or y_2 or y_3.

Def: If $f: A \to B$ is a function, we call A the domain of f and B the codomain of f.

If $f(a) = b$, b is the image of a, and a is the preimage of b.

Def: If $f: A \to B$ is a function, the image or range of a is the set of all images of all elements of A. Denoted $f(A)$.



$f: A \to B$ is a function.

This tells us $f(1) = a$, $f(2) = a$, $f(3) = c$, $f(4) = d$, and $f(5) = b$.

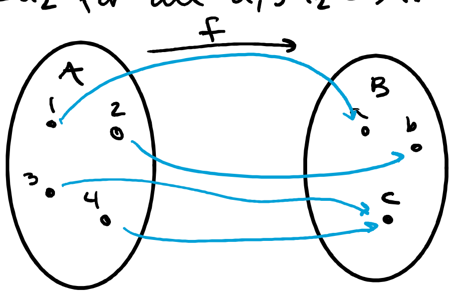
Def: Let $f_1: A \to \mathbb{R}$ and $f_2: A \to \mathbb{R}$ be functions. Then

1) $f_1 + f_2 : A \to \mathbb{R}$ is a function
   $(f_1 + f_2)(x) = f_1(x) + f_2(x)$
2) $f_1 \cdot f_2 : A \to \mathbb{R}$ is a function
   $(f_1 \cdot f_2)(x) = f_1(x) \cdot f_2(x)$

If $f_1: \mathbb{R} \to \mathbb{R}$ is $f_1(x) = x^2 - 1$ and $f_2: \mathbb{R} \to \mathbb{R}$ is $f_2(x) = -x^2 - 2$ find $f_1 + f_2 : \mathbb{R} \to \mathbb{R}$ and $f_1 \cdot f_2 : \mathbb{R} \to \mathbb{R}$

Def: Let $f: A \to B$ and let $S \subseteq A$. Then the image of S is a subset of B containing all images of all elements in S. Denoted $f(S) = \{ b \mid \exists s \in S (b = f(s)) \}$

Def: A function $f: A \to B$ is <u>one-to-one</u> or <u>injective</u> if $f(a_1) = f(a_2)$ implies $a_1 = a_2$ for all $a_1, a_2 \in A$. (same $y \Leftrightarrow$ same x)

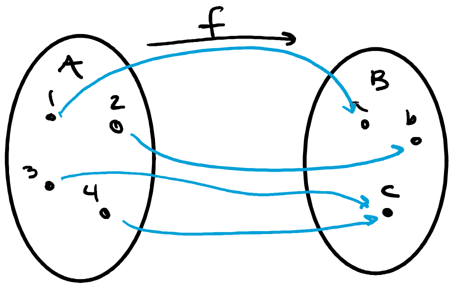


This f is <u>not</u> one-to-one because $f(3) = c = f(4)$ but $3 \neq 4$.

Show $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x + 1$ is one-to-one.

Let x & y be any two elements of $\mathbb{R}$. Suppose $f(x) = f(y)$. Then $x + 1 = y + 1$. Then $x = y$. Therefore f is a one-to-one function.

Def: A function $f: A \to B$ is <u>onto</u> or surjective if for every $b \in B$ there is at least one $a \in A$ such that $f(a) = b$.



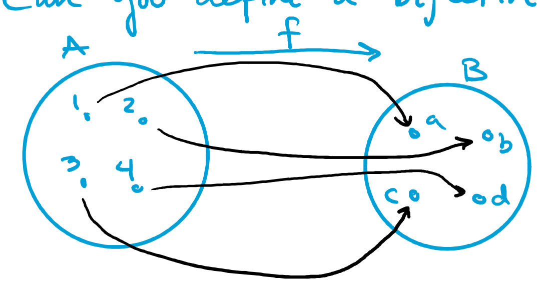
$f: A \to B$ is onto since every element in B is the image of an element of A.

Show $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x+1$ is surjective.

Let $y \in \mathbb{R}$ be any element. Then f is surjective if there is an $x \in \mathbb{R}$ such that $f(x) = y$. Then $x+1 = y$. Then $x = y-1$, and $y-1 \in \mathbb{R}$ so f is surjective.

Def: If $f: A \to B$ is both one-to-one and onto, it is bijective.

Can you define a bijective function?



Summary of Strategies

Show injective: Assume $f(x) = f(y)$ for arbitrary x, y. Show $x = y$.

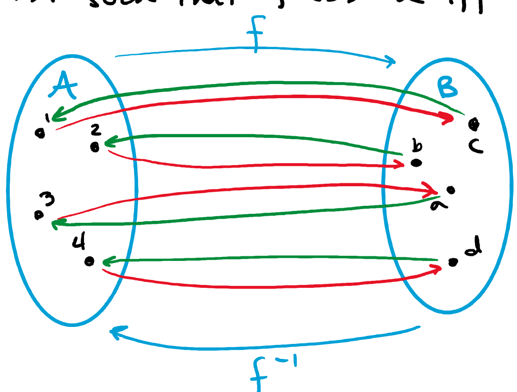
Show not injective: Find special case where $f(x) = f(y)$ but $x \neq y$.

Show surjective: Let $y \in B$ be arbitrary. Find $x \in A$ s.t. $f(x) = y$

Show surjective: Let $y \in B$ be arbitrary. Find $x \in A$ s.t. $f(x) = y$.

Show not surjective: Find special case of $y \in B$ where there is no x s.t. $f(x) = y$.

Definition: If $f: A \to B$ is a bijective function, the inverse function of f is the function $f^{-1}: B \to A$ such that $f^{-1}(b) = a$ iff $f(a) = b$.



Definition: A bijective function is invertible since we can define the inverse function.

Define a function $f: \{1,2,3\} \to \{a,b,c\}$ which is invertible. What is the inverse?

$$\begin{cases} f(1) = a \quad f(2) = b \quad f(3) = c \\ f^{-1}(a) = 1 \quad f^{-1}(b) = 2 \quad f^{-1}(c) = 3 \end{cases}$$

Definition: Let $g: A \to B$ and $f: B \to C$ be functions. The composition of f and g a function denoted $f \circ g: A \to C$ defined as $\quad (f \circ g)(a) = f(g(a))$

Let $g: \{a,b,c\} \to \{1,2,3\}$ and $f: \{1,2,3\} \to \{a,b,c\}$ where

$$\begin{cases} g(a) = 2 \quad g(b) = 3 \quad g(c) = 1 \\ f(1) = b \quad f(2) = c \quad f(3) = a \end{cases}$$

What is the composition $f \circ g$?

What is the composition $g \circ f$?

$f \circ g$: $(f \circ g)(a) = f(g(a)) = f(2) = c$

$(f \circ g)(b) = f(g(b)) = f(3) = a$

$(f \circ g)(c) = f(g(c)) = f(1) = b$

$(f\circ g)(b) = f(g(b)) = f(3) = a$

$(f\circ g)(c) = f(g(c)) = f(1) = b$

$\underline{g\circ f}$: $(g\circ f)(1) = g(f(1)) = g(b) = 3$

$(g\circ f)(2) = g(f(2)) = g(c) = 1$

$(g\circ f)(3) = g(f(3)) = g(a) = 2$

<u>Definition</u>: The <u>identity function</u> $\iota : A \to A$ is defined by $\iota(a) = a$.

<u>Note</u>: For functions $g: A\to B$ and $f: B \to A$ if $f\circ g : A \to A$ is the identity function then $g = f^{-1}$ and $f = g^{-1}$.

In this case, $f\circ g = \iota_A : A \to A$ while $g \circ f = \iota_B : B \to B$.

<u>Definition</u>: The <u>graph</u> of a function $f: A \to B$ is the set of ordered pairs

$$\{(a, f(a)) \mid a \in A\} \subseteq A \times B$$

Works the same as you're used to <u>but</u> be careful of what your domain and codomain are!

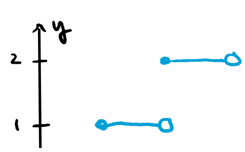
Special Functions

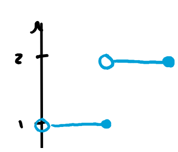
<u>Definition</u>: The <u>floor function</u> assigns to the real number x, the largest integer less than or equal to x.

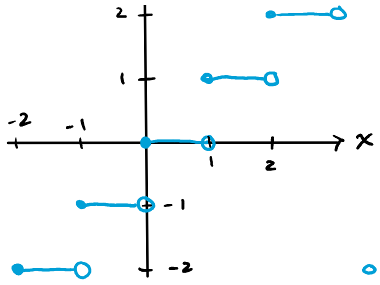
$\lfloor x \rfloor$ or $\lfloor \cdot \rfloor : \mathbb{R} \to \mathbb{Z}$ $\quad (\lfloor 3.104 \rfloor = 3)$

<u>Definition</u>: The <u>ceiling function</u> assigns to the real number x, the smallest integer greater than or equal to x.

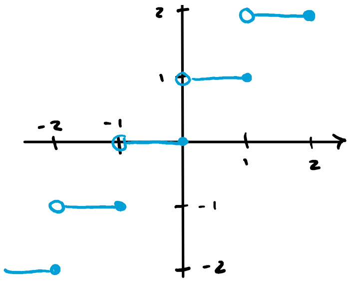
$\lceil x \rceil$ or $\lceil \cdot \rceil : \mathbb{R} \to \mathbb{Z}$ $\quad (\lceil 3.104 \rceil = 4)$



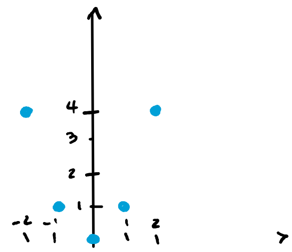




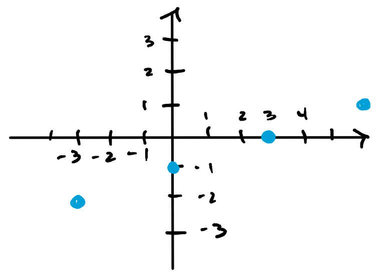
$y = \lfloor x \rfloor$



$y = \lceil x \rceil$



$y = x^2 : \mathbb{Z} \to \mathbb{Z}$



$y = \frac{1}{3}x - 1 : \mathbb{Z} \to \mathbb{Z}$

Definition: The factorial function assigns to the natural number n, the product $1 \cdot 2 \cdot 3 \cdots (n-1) \cdot n$. Denoted

$n!$ or $\cdot ! : \mathbb{N} \to \mathbb{Z}^+$

$0! = 1$, $1! = 1$, $6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$

Definition: A partial function $f : A \to B$ is a function with an assigment of elements of B to only some (a subset) of A. This subset is called the domain of definition. The partial function is undefined at elements not in the domain of definition.

Definition: If the domain of definition of a function $f : A \to B$ is all of A, then f is a total function.

$f : \mathbb{N} \to \mathbb{N}$		$f : \mathbb{R} \to \mathbb{R}$
$f(n) = \sqrt{n}$	vs.	$f(x) = \sqrt{x}$
is total function		is partial function
		domain of def $= [0, \infty)$